

Students are tasked with developing a web application for the "UTAR Library Portal. This application should encompass user registration, profile editing, and library information features. Your objective is to create a seamless user experience, guiding users effortlessly from the homepage through the registration process and into the profile management system. Ensuring accurate data collection and robust validation is crucial. Here are the essential components to include in your project:

Task 1. Homepage:

[10 Mark]

- Begin by designing a homepage that includes the UTAR logo, as shown in Figure 1. (2 Marks)
- Add the header text “Welcome to UTAR Library Portal” below the header, Figure 1. (2 Marks)
- Include three distinct images representing registration, profile editing, and library details, Figure 1. (3 Marks)
- Create links that direct users to the registration form page, the edit profile page, and the Details page, Figure 1. (3 Marks)

Notes:

- Students should refer to the images provided in the exam question folder to design the UI.

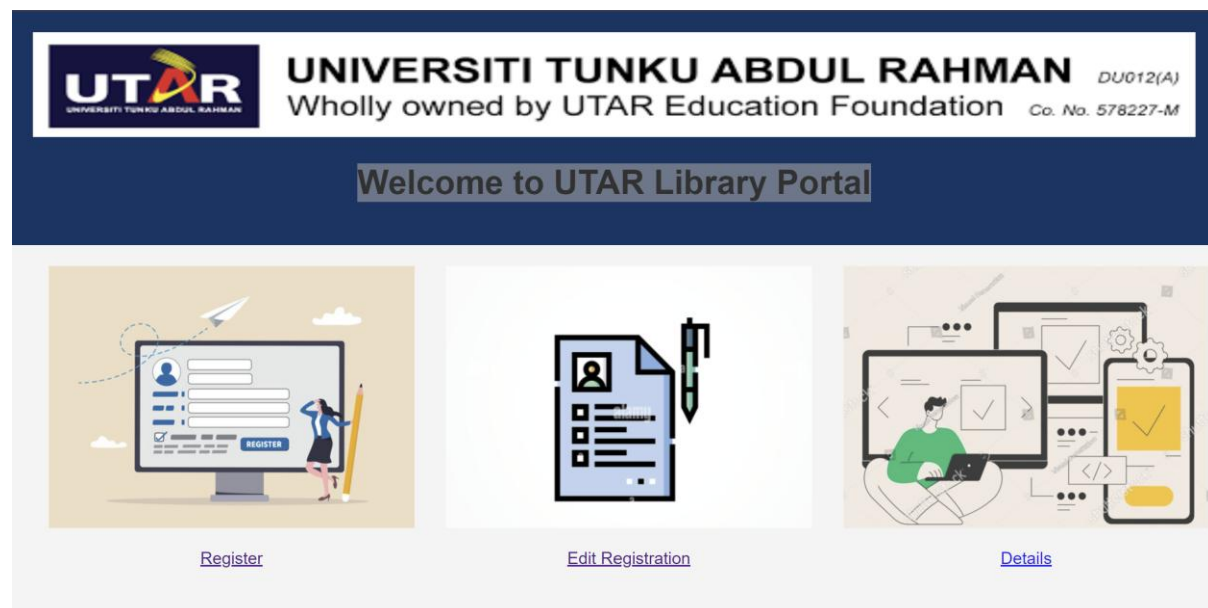


Figure 1 - Home Page

Task 2. Registration Form Development:

[14 Marks]

Build the registration form page where staff/students can sign up for the library. Here are the essential components to include in your project:

- When users click on the link that developed in the task 1 for registration, section (d), it navigates them into the registration page. (2 Mark)

b) Follow the CSS styling of the sample form in the following figure.

(10 Marks)

The image shows a registration form titled "Create a New Account". The form is centered on a light gray background. It contains the following fields and elements:

- Name:** A text input field.
- Email:** A text input field.
- Student/Staff ID:** A text input field.
- Department:** A dropdown menu with "Department of Electrical and Electronic Engineering" selected.
- Password:** A text input field.
- Register:** A dark blue button with white text.

Figure 2 - Registration Form

- c) Start by adding fields to capture essential information such as the staff/student's name and email address, ID, and password. (2 Marks)
- d) Create a dropdown menu (select box) offering department's name as a selection of three distinct research topics: (1 Marks)
- Department of Electrical and Electronic Engineering
 - Department of Internet Engineering and Computer Science
 - Department of Mathematics

Task 3. client-side validation:

[10 Marks]

- a) Implement client-side validation to ensure users provide accurate and complete information before submitting the form. Validation criteria include:
- No empty fields. (each 1 Marks)
 - Email address will have two different validations: 1- not empty field which is mentioned above, 2- following the standard format. (2 Marks)
 - Password must be more than 4 characters. (1 Marks)
- b) Enhance the user experience by providing clear error messages below each field to guide users through the form with a red colour. (2 Marks)

Task 4. Database Configuration and Scripting for registration:

[11 Marks]

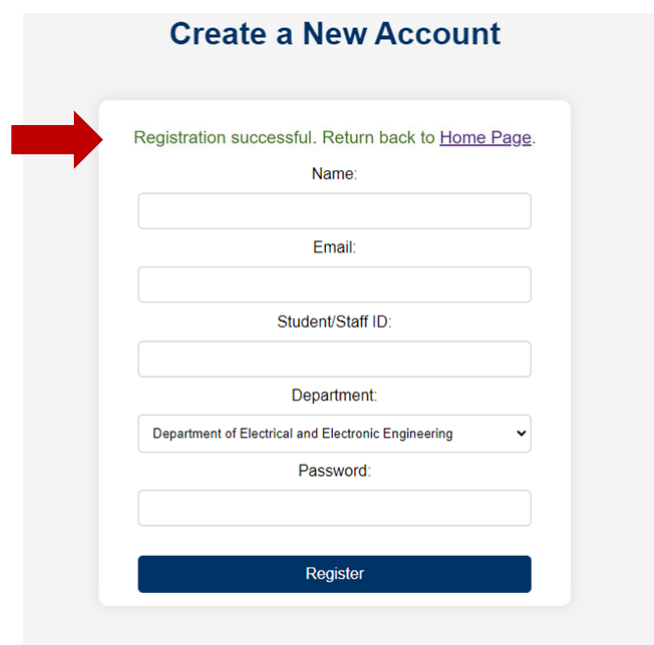
- a) Create a database named "utar_db," with the collation set to "utf8_general_ci." (1 Marks)

- b) Write SQL statements to create a table named "utar_table" within the "utar_db" database, with the structure outlined in the following table. (3 Marks)

Attribute	Tyep	Key
ID	INT(11) AUTO INCREMENT	Primary
Name	VARCHAR(255)	
Email	Text	Unique
Staff/Student ID	Text	
Department	Text	
Password	Text	

Note:

- Ensure to write the SQL CREATE DATABASE and TABLE statements and save them in a file named "db.sql." Creating database and table in the "phpMyAdmin" is not acceptable.
 - Adding the code and screen shot of the database and table in phpMyAdmin is required.
- c) Store all the information that users input in the registration form created in task 2 into the "utar_library" table in the "utar_db" database. (5 Marks)
- d) On successful registration, a message like following figure should be shown. (2 marks)



The image shows a web form titled "Create a New Account". At the top, a green message states "Registration successful. Return back to [Home Page](#)." Below this, the form contains input fields for "Name:", "Email:", "Student/Staff ID:", "Department:" (with a dropdown menu showing "Department of Electrical and Electronic Engineering"), and "Password:". A blue "Register" button is at the bottom. A red arrow points to the success message.

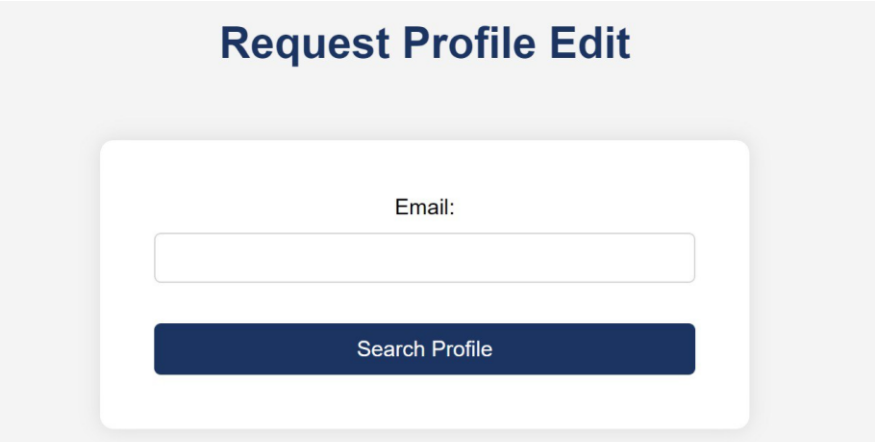
Figure 3 - Successful registration

Task 5. Database Scripting for edit profile:

[10 Marks]

Build Update Profile page where staff/students can edit their profile. Here are the essential components to include in your project:

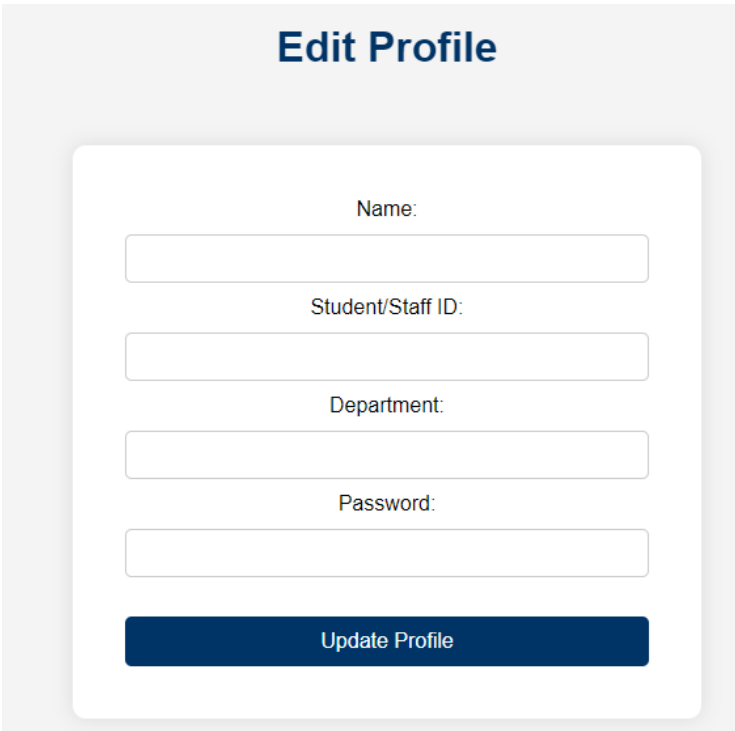
- a) When the user clicks on "Edit Profile," from the Home Page, they are navigated to a page containing a form designed to capture the ID of the user whose profile needs to be edited. (3 Marks)



The image shows a web form titled "Request Profile Edit" in a dark blue font. The form is contained within a white rounded rectangle with a subtle shadow. It features a label "Email:" above a single-line text input field. Below the input field is a dark blue button with the text "Search Profile" in white.

Figure 4 - Request Profile Edit

- a) If the profile is found in the database, the form shown in the following figure will be shown to the user to key in the information again and save the changes into db. The db must be updated when the update profile button is clicked. (5 Marks)



The image shows a web form titled "Edit Profile" in a dark blue font. The form is contained within a white rounded rectangle with a subtle shadow. It features four labels with corresponding text input fields: "Name:", "Student/Staff ID:", "Department:", and "Password:". At the bottom of the form is a dark blue button with the text "Update Profile" in white.

Figure 5 - Edit Profile

- b) Upon successful edit operation, the page should automatically redirect the user to the "Home Page." (1 Marks)
- c) If the profile edit is unsuccessful, the user should remain on the current page, and an error message should be displayed, indicating the issue. (Design of error message is optional.) (1 Marks)

Task 6: Details page:

[5 Marks]

Build the details page where users can see the details information of the library. When the user clicks on "Details," they are navigated to a page containing the details information. The text is provided in the question folder.